String Theory is a theory of Quantum Gravity

Our focus will be on the perturbative approach to the Bosonic String & show why this gives a consistent theory of Quantum gravity.

The classical Gravity is defined completely by the Einstein's field eq^n

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}$$

This eq^n is derived by the variation of the classical action called as the "Einstein-Hilbert Action"

$$S_{EH} = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g}\, R$$

$g \rightarrow \det(g_{\mu\nu})$, $R \rightarrow$ Ricci Scalar

$G \rightarrow$ Gravitational Constant.

Then the factor '$8\pi G$' can be related to the reduced Planck's mass 'M_p' as

$$M_p = \sqrt{\frac{\hbar c}{8\pi G}} = 4.34 \times 10^{-9}\ Kg$$

$$\rightarrow \quad 8\pi G = \frac{\hbar c}{M_p^2}$$

$$\Rightarrow R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{\hbar c}{M_p^2 c^4} T_{\mu\nu}$$

$E_{Planck} = M_p c^2$ is the reduced planck energy given by 2.4×10^{18} GeV.

$$\rightarrow R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{\hbar c}{E_{Planck}} T_{\mu\nu}$$

In the units of $\hbar = c = 1$, we get

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{T_{\mu\nu}}{E_{Planck}}$$

Thus in natural units,

$$M_p = 2.4 \times 10^{18} \text{ GeV}$$

Planck's mass represents a Hypothetical particle which due to its mass in a tiny Space has enough density to be a 'Black Hole'.

In String theory, Planck's mass sets an energy scale for the Scenario of when the String theory becomes important.

The quantum effects becomes important when we work in the energy scale comparable to Plank's mass.

For more explanation read Polchinski & David Tong's lecture notes.

Throught these upcoming String notes we shall use the Space-time of the total 'D' dimensions. '1' time like coordinate with index '0' & 'D-1' Space-like coordinates with indices from '1 to D-1'.

The metric Signature we use is given by the D dimensional minkowski metric

$$\underline{\eta_{\mu\nu} = \mathrm{diag}(-,+,+,---,+)}$$

Note that this is opposite to my Quantum field theory notes.

Now, we shall look into the dynamics of a relativistic free point Particle.

Relativistic Point Particle

In D dimensional space-time in a Minkowski space of $\mathbb{R}^{1,D-1}$, the action of a relativistic point particle is given by,

$$S = -m \int dt \sqrt{1 - \left|\frac{d\vec{x}}{dt}\right|^2}$$

where $\vec{x}(t) = [x^1(t), x^2(t), \cdots x^{D-1}(t)]$ is the location of the particle at any time 't'. 'm' is the mass of the particle.

$$\frac{d\vec{x}}{dt} = \dot{\vec{x}} = \text{spatial velocity.}$$

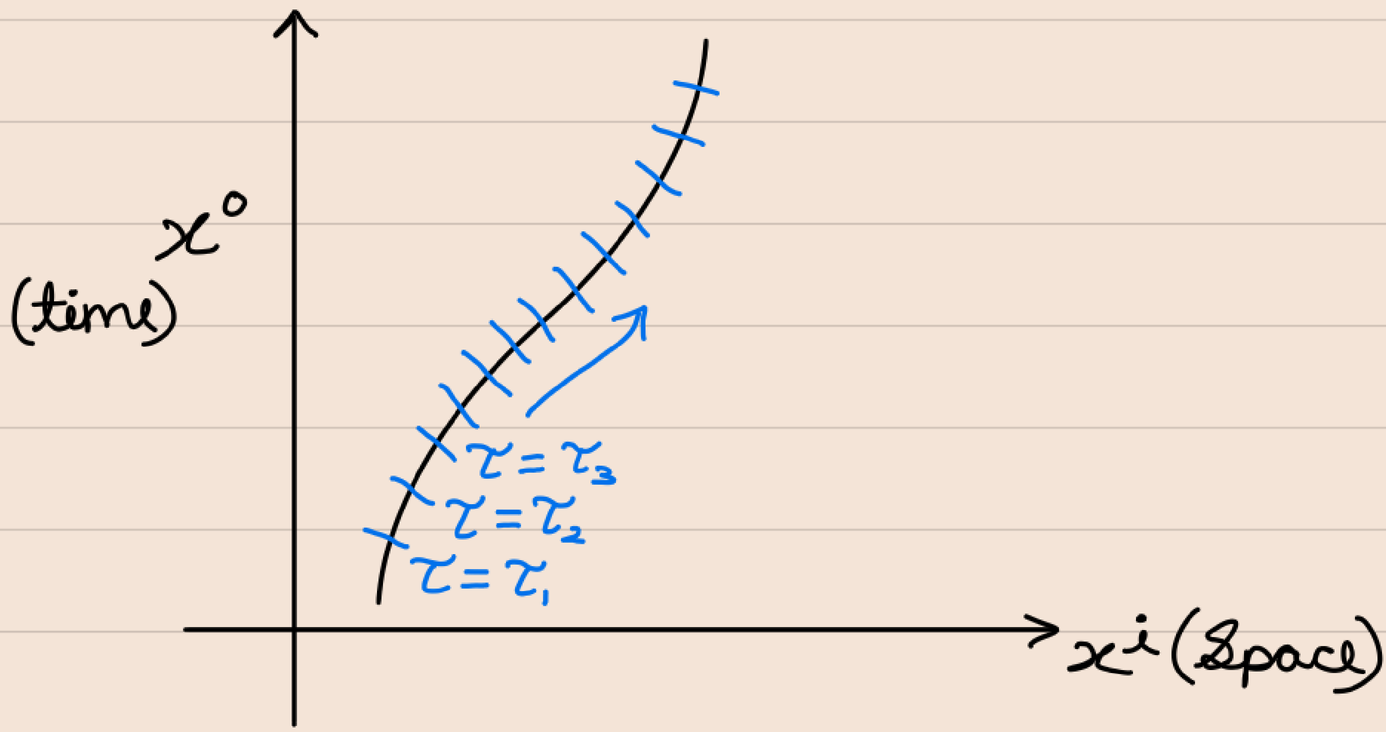
This gives its spatial momentum as,

$$\vec{p} = \frac{m\dot{\vec{x}}}{\sqrt{1 - |\dot{\vec{x}}|^2}}$$

& Energy satisfying $\underline{E^2 = m^2 + |\vec{p}|^2}$

The previous action treats time 't' Seperately compared to Space, where time is not a degree of freedom.

But if we introduce a parameter 'τ' which labels the position of the particle in the D dim. Space time, then every coordinate 'X^μ' can be parameterized by 'τ'.



We can parameterize every Space-time coordinate by 'τ'.

$$\Longrightarrow X^\mu \longrightarrow X^\mu(\tau)$$

Note that this parameterization has a redundancy. If $\tau \rightarrow \bar{\tau}$ as a new parameter related by some monotonic function, still the theory remains the same. Then, under this reparameterization, $X^{\mu}(\tau) \longrightarrow \bar{X}^{\mu}(\bar{\tau})$ but the world line described by both '$X^{\mu}(\tau)$' and '$\bar{X}^{\mu}(\bar{\tau})$' will be exactly the same. This is '<u>Reparameterization Invariance</u>'

So a simple 'Poincare Invariant' (under $X^{\mu} \xrightarrow{\text{Poincare}} \Lambda^{\mu}_{\ \nu} X^{\nu} + a^{\mu}$') action is given by,

$$S = -m \int d\tau \sqrt{-\dot{X}^{\mu} \dot{X}^{\nu} \eta_{\mu\nu}}$$

The interpretation is, τ is the proper time of the particle. And $\dot{X}^{\mu} = \dfrac{dX^{\mu}(\tau)}{d\tau}$ is the Four velocity.

$\Rightarrow$ 'S gives four distance'

Even though $X^0(\tau)$ looks like it is a dynamical variable, it is not. Because, consider reparameterization

$$\tau \longrightarrow \bar{\tau} = \bar{\tau}(\tau) \longrightarrow \dot{X}^\mu = \left(\frac{dX^\mu}{d\bar{\tau}}\right)\left(\frac{d\bar{\tau}}{d\tau}\right)$$

$$\Rightarrow S = -m\int d\tau \left(\frac{d\bar{\tau}}{d\tau}\right)\sqrt{\left(\frac{dX^\mu}{d\bar{\tau}}\right)\left(\frac{dX^\nu}{d\bar{\tau}}\right)\eta_{\mu\nu}}$$

$$\Longrightarrow S = -m\int d\bar{\tau}\sqrt{\left(\frac{dX^\mu}{d\bar{\tau}}\right)\left(\frac{dX^\nu}{d\bar{\tau}}\right)\eta_{\mu\nu}}$$

Thus action is invariant under this reparameterization.

Now we simply set $\tau = X^0(\tau) = t$, then we recover the previously defined action of '$-m\int dt\sqrt{1-\left|\frac{d\vec{x}}{dt}\right|^2}$'

$\longrightarrow X^0(\tau)$ is not a true degree of freedom & the number of true degrees of freedom is still the usual 'D-1'.

Also identifying $P^\mu = m\dfrac{dX^\mu}{d\tau}$ as the four momentum, we get a constraint,

$$p^\mu p^\mu + m^2 = 0$$

$$\rightarrow \underline{(P^0)^2 = E^2}$$

This is a constraint of the system. A particle has to satisfy this. From the world line perspective, particle has to keep moving in the time-like direction. It cannot be still in space-time.

But still, the downside of this theory is, it doesn't consider the massless particles. So we should further modify the action, to include the dynamics of the massless particle. This can be achieved using a auxillary field called 'Einbein'.

Consider the action given by

$$S = \frac{1}{2}\int d\tau \left[\varepsilon^{-1}\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu} - \varepsilon m^2\right]$$

where $\varepsilon \to \varepsilon(\tau)$ is an auxillary 'Einbein' field. This action has the usual Poincare invariance. We also need it to be reparameterization invariant where we require '$\underline{\varepsilon(\tau)d\tau = \bar{\varepsilon}(\bar{\tau})d\bar{\tau}}$'.

'$\varepsilon(\tau)$' is actually related to the world-line metric '$\gamma_{\tau\tau}(\tau)$' as '$\underline{\varepsilon(\tau) = \sqrt{-\gamma_{\tau\tau}(\tau)}}$'.

By introducing the Einbein field, we have increased the degrees of freedom by one.

The eq^n of motion for X^{μ} is,

$$\frac{dP^{\mu}}{d\tau} = 0 \quad , \text{ where } P^{\mu} = \varepsilon^{-1}\dot{X}^{\mu}$$

$$\Rightarrow \quad \underline{\ddot{X}^{\mu} = \varepsilon^{-1}\dot{\varepsilon}\dot{X}^{\mu}}$$

& Eq^n of motion for $\varepsilon(\tau)$ is

$$\varepsilon^{-2}\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu}+m^2=0$$

Thus the $\varepsilon(\tau)$ is completely fixed.

$$\rightarrow \quad p^{\mu}=\varepsilon^{-1}\dot{X}^{\mu}=\frac{m\dot{X}^{\mu}}{\sqrt{-\dot{X}^{\rho}\dot{X}^{\sigma}\eta_{\rho\sigma}}}$$

$$\longrightarrow \quad \underline{p^{\mu}p^{\nu}\eta_{\mu\nu}=-m^2} \Longrightarrow \text{Obey's mass-shell}$$

Using $\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu}+\varepsilon^2 m^2=0$ in the action, we get,

$$S=\frac{1}{2}\int d\tau\left[\varepsilon^{-1}\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu}-\varepsilon m^2\right]$$

$$=\frac{1}{2}\int d\tau\,\varepsilon\left[\varepsilon^{-2}\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu}-m^2\right]$$

$$=\int d\tau\,\varepsilon\left[\varepsilon^{-2}\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu}\right]$$

$$= \int d\tau \, \frac{\sqrt{-\dot{X}^\sigma \dot{X}^\rho \eta_{\sigma\rho}}}{m} \, \underbrace{\left[\epsilon^{-2} \dot{X}^\mu \dot{X}^\nu \eta_{\mu\nu}\right]}_{-m^2}$$

$$\rightarrow \underline{S = -m \int d\tau \sqrt{-\dot{X}^\mu \dot{X}^\nu \eta_{\mu\nu}}}$$

Thus the action with $\epsilon(\tau)$ describes same physics independent of $\epsilon(\tau)$.

Now, for massless particles,

$$S = \frac{1}{2} \int d\tau \, \epsilon^{-1} \dot{X}^\mu \dot{X}^\nu \eta_{\mu\nu}$$

& the eq^n of motion for X^μ is,

$$\frac{dP^\mu}{d\tau} = 0 \quad , \text{ where } P^\mu = \epsilon^{-1} \dot{X}^\mu$$

Eq^n of motion for $\epsilon(\tau)$: $\epsilon^{-2} \dot{X}^\mu \dot{X}^\nu \eta_{\mu\nu} = 0$

$$\rightarrow \underline{P^\mu P^\nu \eta_{\mu\nu} = 0} \rightarrow \text{On-mass shell Conditions for the massless particles.}$$

Thus the action given by,

$$S = \frac{1}{2}\int d\tau \left[\varepsilon^{-1}\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu} - \varepsilon m^2\right]$$

describes a relativistic free particle (massive & massless), but with the extra Gauge Symmetry where we have the freedom to fix one of $X^{\mu}(\tau)$ at will, because of reparameterization invariance.

Eg: Proper time gauge fixes $X^{0}(\tau)$ using, $(d\tau)^2 = (dX^0)^2 + (d\vec{X})^2$

$$\longrightarrow dS = -\int d\tau$$

And the Enbein field is not independent, it is given by the Constraint,

$$\varepsilon^{-2}\dot{X}^{\mu}\dot{X}^{\nu}\eta_{\mu\nu} + m^2 = 0$$

Moreover, this action is quadratic & thus easily quantizable.